

Om CHAUHAN

602-807-2203 | omrajesh9203@gmail.com | linkedin.com/in/omchauhan9203 | github.com/OmChauhan9

SUMMARY

Software Engineering M.S. candidate with experience building **backend, cloud, and AI/ML systems** across **Spring Boot, AWS/GCP, Docker, Terraform, PyTorch, LangGraph, and RAG**. Developed production-oriented projects in **multi-agent AI, computer vision, edge inference, autoscaling infrastructure, and CI/CD automation**.

EDUCATION

Master of Science in Software Engineering

Arizona State University, Tempe, AZ

Aug 2024 – May 2026

GPA: 3.78/4.0

Bachelor of Engineering in Computer Engineering

Gujarat Technological University, India

Aug 2020 – May 2024

GPA: 3.39/4.0

EXPERIENCE

Software Engineer Intern, Ahmedabad, India | Axisray

June 2023 – June 2024

- Engineered a fullstack HR automation system using **Spring Boot** and **MySQL**, reducing manual HR operations for a **50+ person** team.
- Built a voice-driven HR query interface by integrating **Amazon Alexa Skills Kit** with **AWS Lambda**, enabling employees to retrieve HR information hands-free through natural language requests.
- Automated the build and deployment pipeline using **AWS CodePipeline, CodeBuild, CodeDeploy, and CodeArtifact**, triggered via **GitHub** webhooks, reducing deployment errors to near-zero.
- Developed and trained a real-time fire detection system from scratch using **YOLOv5** on a custom dataset of **10,000 images and 100 videos**, achieving **95% accuracy** on a held-out test set with a **40% reduction** in false positive alerts.

PROJECTS

SafeGuard: Mobile SOS Application | *Node.js, Socket.io, AWS, Terraform, Docker*

- Architected and migrated AWS infrastructure from **ECS Fargate, ALB, and ElastiCache** to a single **EC2** instance with **Docker Compose**, cutting projected monthly cloud spend from **~\$50 to ~\$4/month** for a **92% reduction**.
- Implemented multi-phase location tracking logic in **Node.js** and **Socket.io** handling active SOS triggers, passive background watching, wake-on-push for killed app processes, and offline dead-zone protocol.
- Authored modular **Terraform** infrastructure for **VPC, ECS, ElastiCache Redis, Cognito, and Secrets Manager** with **GitHub Actions CI/CD** using **OIDC authentication**, eliminating long-lived AWS credentials.

StratusVault: Secure Cloud File Share on GCP | *Spring Boot, GCP, PostgreSQL, Firebase Auth*

- Designed and implemented a cloud-native document storage service on **GCP** using **Cloud Run** for the stateless **Spring Boot API**, **Cloud SQL PostgreSQL** for relational metadata, and **Cloud Storage** for private object storage.
- Built a file-level ACL system using **Cloud SQL** and **Firebase Authentication**, verifying Firebase ID tokens server-side and resolving per-file owner and shared-user permissions from the database with no public **GCS** bucket exposure.

Hybrid Cloud & Edge AI System | *AWS, EC2, SQS, Lambda, IoT Greengrass, PyTorch*

- Built a queue-depth autoscaling controller that dynamically provisioned and terminated up to **15 EC2 app-tier instances** based on **SQS** backlog, scaling from zero under idle conditions and achieving sub **1.2s average latency** across a **100-request benchmark**.
- Architected an edge computing pipeline using **AWS IoT Greengrass** and **MQTT** to run **MTCNN** face detection locally on an **EC2 IoT client**, transmitting only detected face crops to **SQS** for cloud-side **FaceNet** recognition.
- Deployed a serverless **AWS Lambda** inference backend with **SQS queues, S3 persistence**, and containerized **PyTorch** models, achieving **100% benchmark accuracy**.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, C#, PL/SQL, SQL, JavaScript/TypeScript, HTML, CSS.

Frontend: ReactJS, React Native (Expo), NextJS, Angular, Redux, React Query, jQuery, Bootstrap.

Backend & Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Cloud, Hibernate, ASP.NET Core, .NET, Entity Framework Core, Flask, REST APIs, GraphQL, Socket.IO, JUnit.

Cloud & Databases: AWS (EC2, S3, Lambda, Step Functions, SQS, API Gateway, SES, IoT Greengrass), GCP (Cloud Run, Pub/Sub, Cloud Spanner, Cloud Storage), Azure, Firebase, MongoDB, Redis, DynamoDB, MySQL, PostgreSQL, SQL Server, SQLite, Oracle.

DevOps & Tools: Docker, Kubernetes, Terraform, CI/CD Pipelines, Git, GitLab, Bitbucket, GitHub Actions, Jenkins, CircleCI, Azure DevOps, Linux, Postman, Swagger, Insomnia, JIRA, Monday, Taiga.

AI/ML & Generative AI: TensorFlow, PyTorch, Hugging Face, Pandas, NumPy, LangChain, LangGraph, LangSmith, Agentic AI, Claude Code, Amazon Kiro, Amazon Q, Cursor AI, Codex.

Coursework: Advanced Data Structures & Algorithms, Cloud Computing, Distributed Database Systems, Operating Systems, Advanced Web Development, Applied Cryptography, Object-Oriented Programming, Computer Architecture.